MACHINE LEARNING • RANDOM FOREST METHODOLOGY

Crop Classification

in Dronten, Flevoland

Multi-temporal Sentinel-2 imagery • Random Forest classifier • 9 crop classes, 4,088 parcels

EuroCrops v2.0

AUTHOR

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PROJECT OVERVIEW

Automated Crop Classification from Satellite Imagery

An end-to-end Random Forest pipeline on Sentinel-2 time-series

WHAT

Automated crop-type identification using machine learning applied to multispectral Sentinel-2 imagery from the intensively farmed Dronten polder region.

WHY

Precision agriculture — *optimise input use parcel-by-parcel*

Yield prediction — *forecast harvests before the season closes*

Environmental monitoring — *track land use change at scale*

Policy support — *independent ground-truth for CAP declarations*

STUDY AREA	SEASON	GROUND TRUTH	MODEL
Dronten, Flevoland	Feb – Sep 2023	EuroCrops v2.0 / BRP	Random Forest + GridSearchCV

Dronten Agricultural Area, Netherlands

STUDY AREA

Dronten, Flevoland — The Netherlands

A young polder, reclaimed from the IJsselmeer in 1968 — now one of Europe's most intensively farmed regions

LOCATION

Flevoland Province, NL
52.5°N, 5.7°E

LAND USE

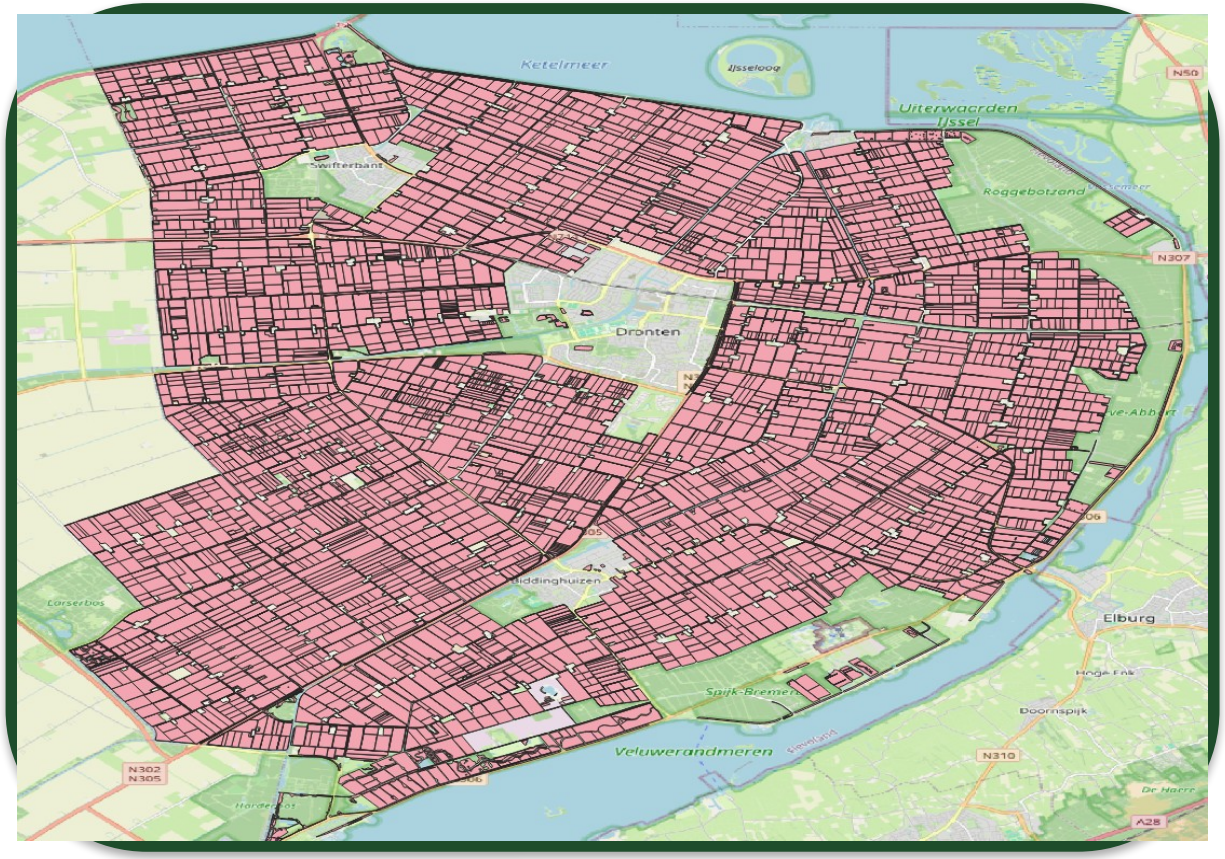
~90% arable farmland
Reclaimed polder land

KEY CROPS

Wheat, Potato, Sugar Beet
Onion, Maize, Oilseeds

TIME FRAME

Growing season 2023
(February – September)



4,088 agricultural parcels across Dronten polder • EuroCrops v2.0

Data source: EuroCrops v2.0 derived from Dutch BRP/PDOK parcel declarations

DATA & GROUND TRUTH

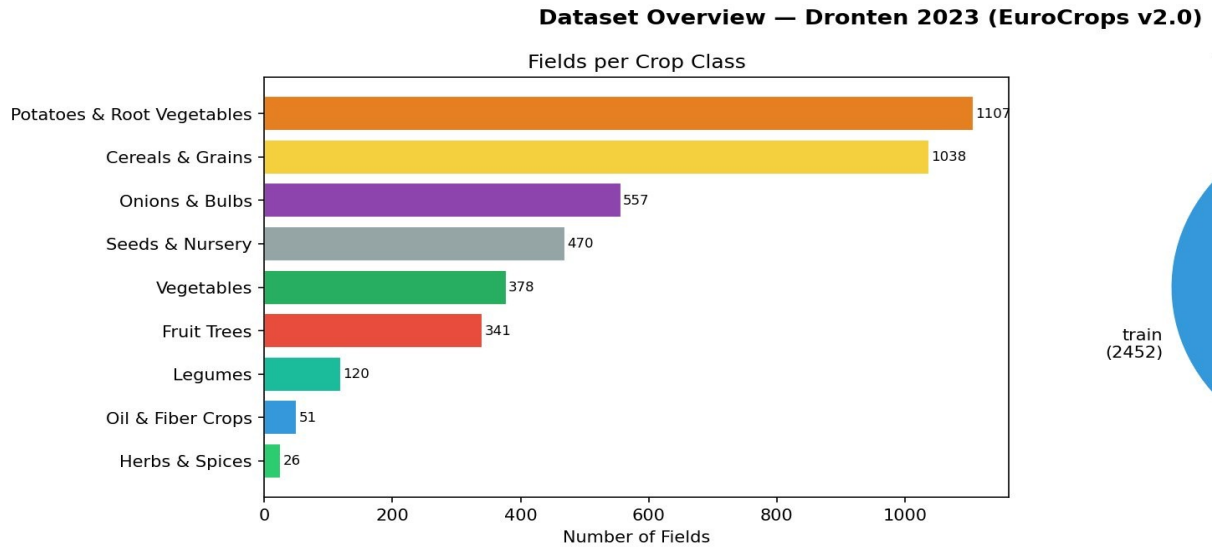
Label Distribution & Class Imbalance

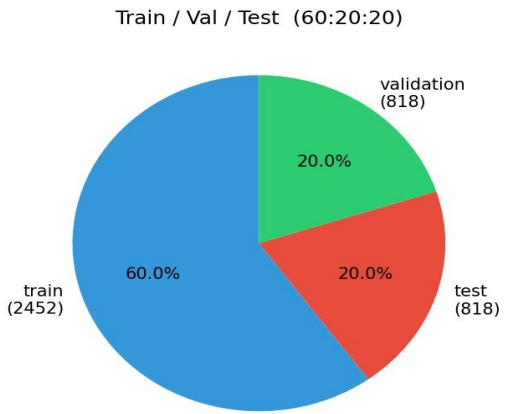
9 crop classes across 4,088 parcels — with a 43:1 imbalance that drives every downstream decision

4,088
parcels

9
crop classes

60 : 20 : 20
train / val / test





CLASS IMBALANCE

Potatoes (1,107) vs Herbs & Spices (26). F1-macro chosen over accuracy to weight all 9 classes equally.

LABEL NOISE

Labels come from farmer CAP subsidy declarations, not field verification — a realistic but noisy benchmark.

Ground truth: EuroCrops v2.0 · BRP/PDOK parcel registrations · 60/20/20 pre-assigned split

WORKFLOW

From Satellite Pixels to Crop Predictions

Five stages — each designed to handle the specific constraints of multispectral time-series data

01

INGEST

Sentinel-2 L2A
via Earth Engine

02

PREPROCESS

SCL cloud masking
Monthly composites

03

EXTRACT

Zonal stats per
parcel polygon

04

ENGINEER

5 spectral indices
× 7 months = 105 feats

05

CLASSIFY

Random Forest
+ GridSearchCV

DESIGN CHOICE

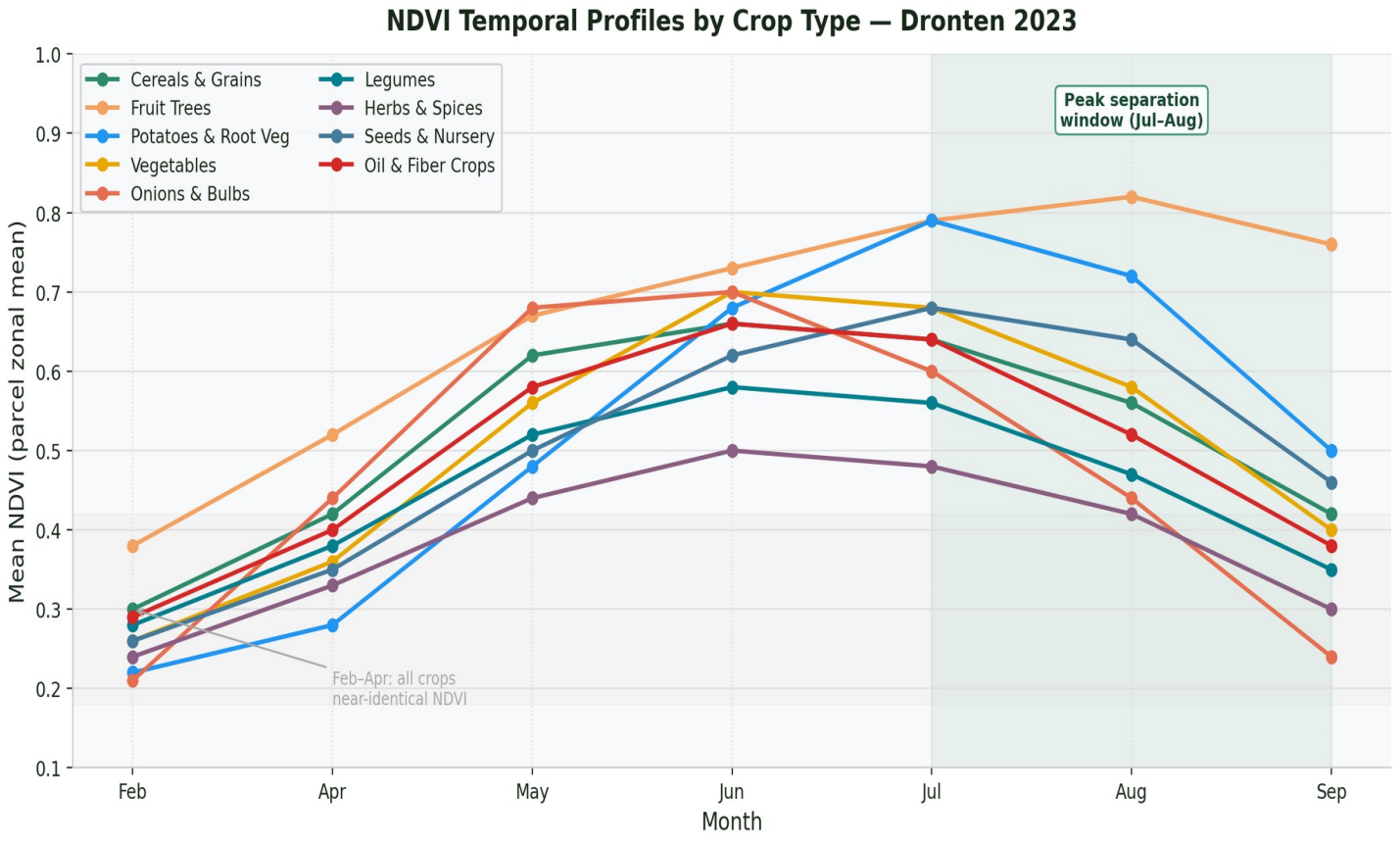
Random Forest was chosen for its native handling of feature collinearity — essential when the same 10 spectral bands are repeated across 7 months, producing 70 correlated raw features plus 35 derived indices.

Full Random Forest methodology follows →

Data, Preprocessing & Feature Engineering

105 Features | **7** Months | **4,088** Parcels | **9** Crop Classes

NDVI Temporal Profiles by Crop Type — Dronten Growing Season 2023



DATA SOURCE & PREPROCESSING

Sentinel-2 L2A SR Harmonized via Google Earth Engine (Feb through Sep 2023).
SCL pixel-level masking -95% scene threshold.
median composite per month - 20 m resolution (EPSG:32631).
Zonal statistics -mean band reflectance per parcel polygon across 4,088 EuroCrops v2.0 fields.
No normalization was applied — RF is invariant to feature magnitude.

FEATURE BREAKDOWN (105 total)

Raw Spectral Bands 10 bands × 7 months

70 features

Spectral Indices NDVI · EVI · NDMI · NDBI · RE-NDVI × 7

35 features

Why no feature selection?
Random Forest handles multicollinearity natively via bootstrapped subspace sampling (~10 features/split).

Method Design — Why RF? Why These Features?

Decisions made BEFORE training — based on problem and data characteristics known upfront

WHY RANDOM FOREST FOR THIS PROBLEM?

01

Correlated features — by design Same 10 bands across 7 months = high multicollinearity before training starts. RF's random subspace sampling (~10 features/split) handles this natively. SVM and linear models break without RFE first.

02

NDVI temporal profiles showed non-linear separation Time series plot showed crops diverging at different months and rates. No straight boundary could separate all 9 classes. RF makes no linearity assumption — proven benchmark in RS crop classification literature.

03

Class distribution showed severe imbalance Class count showed Potatoes (1,107) vs Herbs (26) — 43:1 ratio. A model treating all classes equally ignores rare crops. RF with bootstrapping + class_weight=balanced corrects this without oversampling.

Each index chosen for a specific physical property of crop phenology

WHY THESE 5 SPECTRAL INDICES?

NDVI (B8-B4)/(B8+B4)
Core greenness. Captures full seasonal growth trajectory — the primary signal separating crop types across 7 months.

RE-NDVI (B8-B5)/(B8+B5) **Confirmed #1 RF feature**
Red-edge (705nm) = chlorophyll concentration. Separates broadleaf from narrow-leaf crops at full canopy — where NDVI saturates above 0.8.

EVI 2.5*(B8-B4)/(B8+6B4-7.5B2+1)
Corrects canopy background + atmospheric effects via B2. Keeps discriminating where NDVI saturates — Potatoes vs Fruit Trees at peak season.

NDMI (B8-B11)/(B8+B11)
Leaf water content via SWIR. Onions dry down early, Potatoes sustain moisture. Captures post-harvest divergence NDVI misses.

NDBI (B11-B8)/(B11+B8)
Bare soil / built-up control. Dronten polder fields have drainage canals and bare margins. NDBI flags non-vegetated pixels reducing 20m mixed-pixel noise.

Random Forest Pipeline

Machine Learning Pipeline — Random Forest · Sentinel-2 · Dronten 2023

1 DATA ACQUISITION

Sentinel-2 L2A SR
GEE · COPERNICUS
7 months (2023)
Feb → Sep

2 CLOUD MASKING

SCL pixel-level mask
(classes 4,5,6,7,11)
95% scene threshold
Median composite

3 FEATURE ENGINEERING

10 bands × 7 months
= 70 band features
5 indices × 7 months
= 35 index features

4 DATA SPLITTING

Stratified split
60% train (2,262)
20% val (754)
20% test (754)

5 GridSearchCV TUNING

12 combinations
× 3-fold CV
= 36 fits
f1_macro scoring

6 RANDOM FOREST

150 trees
max_depth = 15
min_split = 5
class_weight balanced

EVALUATION

82.0%	~0.65	~0.77	1.6%
Test Accuracy	F1-macro (9-class equal weight)	Cohen's Kappa (Substantial Agreement)	Val–Test Gap (no overfitting)

RF METHOD -- KALYANI KHAMGAONKAR

Feature Importance Analysis

Top 5 Individual Features

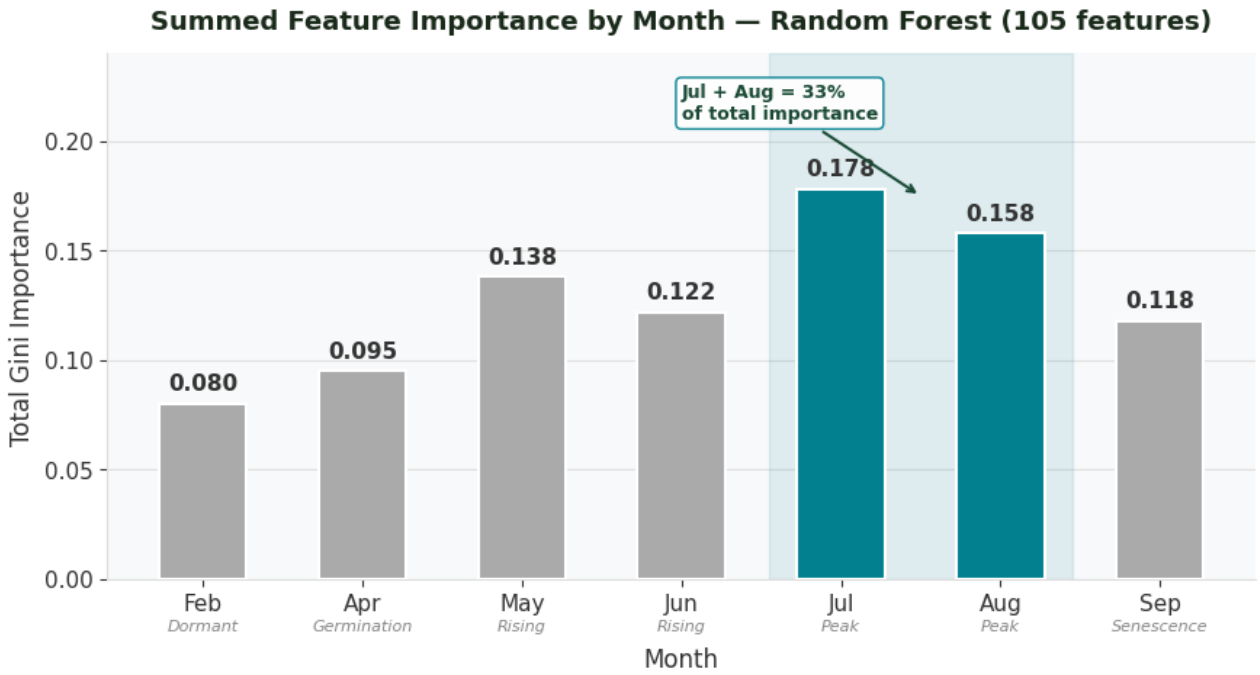
#1 RE_NDVI_Jul 0.0216
#2 B5_Sep 0.0201
#3 EVI_Jul 0.0196
#4 EVI_May 0.0195
#5 NDVI_Jul 0.0195

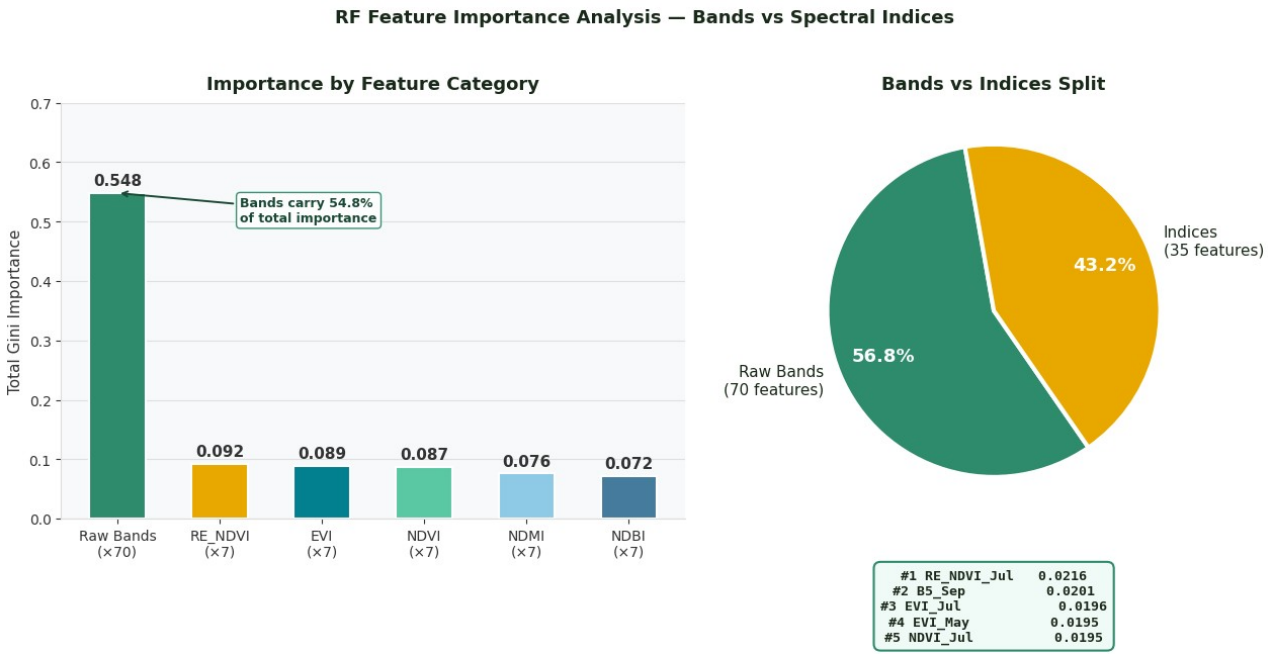
Key Observations

RE_NDVI_Jul dominates (#1)
Red-edge (B5/B8) most sensitive to chlorophyll content differences; peak growing season is the maximum class differentiation window

Raw bands > derived indices
Bands collectively outweigh indices; raw spectral shape contains information beyond normalised ratios (consistent with Esther's RFE retaining 45/70 band features)

Feb importances near zero
Removing Feb/Apr features would reduce dimensionality with negligible accuracy loss

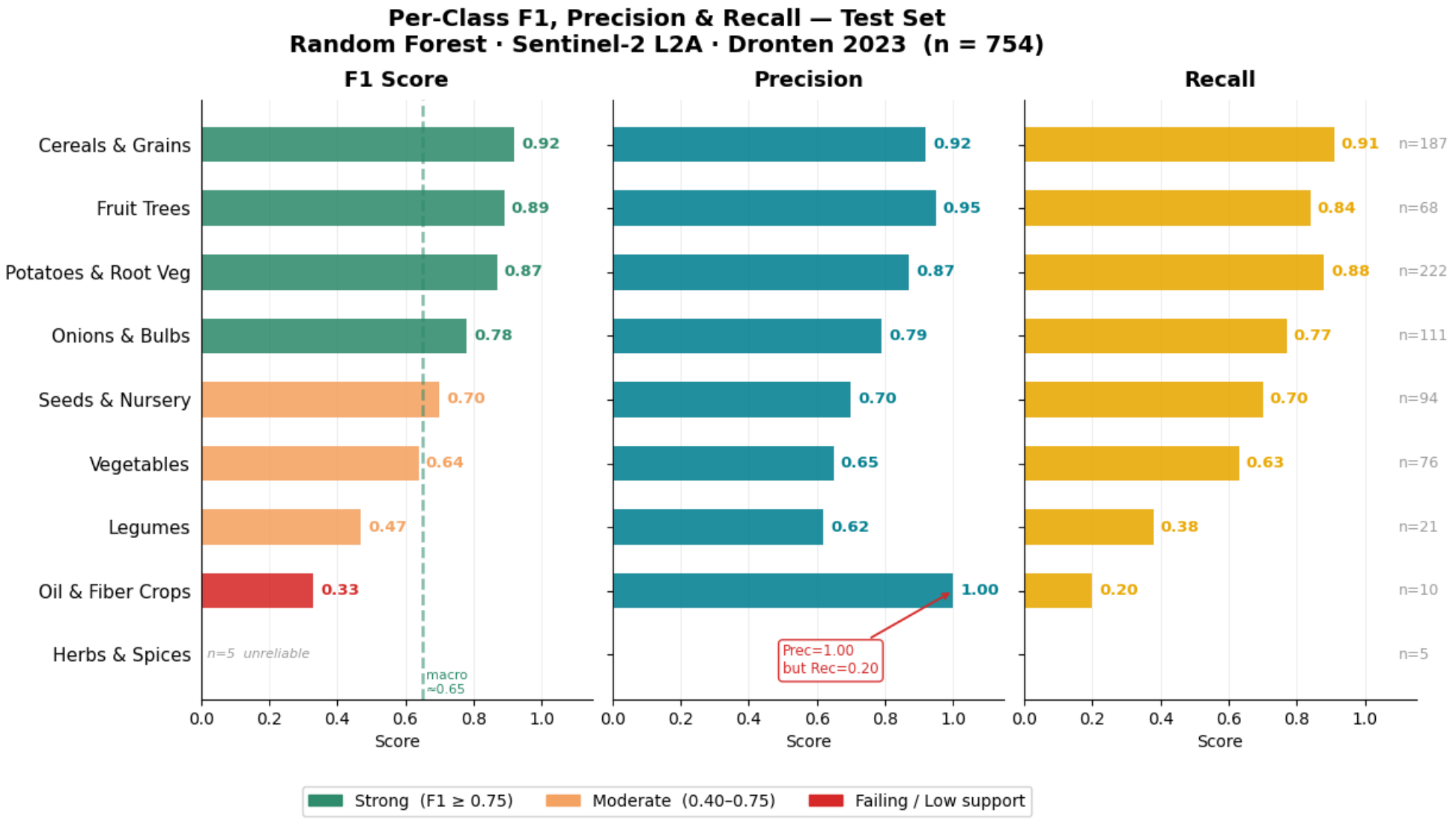




RF METHOD -- KALYANI KHAMGAONKAR

Hyperparameter Tuning & Classification Results

GridSearchCV | n_estimators= 150 | max_depth-15 | min_samples_split-5| max_features--'sqrt'



82.0%
Test Accuracy

~0.65
F1-macro (9-class equal weight)

~0.77
Cohen Kappa (Substantial Agreement)

1.6%
Val-Test gap (no overfitting)

Analysis, Interpretation & Critical Reflection

01 MODEL DESIGN

Why RF? Why these settings?

Bagging = no feature selection needed Each tree sees ~10 random features per split. Correlated NDVI/EVI/NDMI across 7 months handled natively. SVM needed RFE (112 to 70).

max_depth=15 Prevents memorising individual fields. Forces broad phenology patterns. Unlimited depth overfit on Herbs (n=26 train).

f1_macro scoring in GridSearchCV Weights all 9 classes equally. Stops optimiser sacrificing rare classes (Herbs, Oil & Fiber) for majority-class gains.

02 WHAT SCORES MEAN

Decoding every metric

Accuracy 82% vs F1-macro 0.65 — gap is expected Cereals + Potatoes = 54% of test samples. Accuracy rewards majority classes. F1-macro weights all 9 equally — honest metric.

Strong classes — Cereals F1=0.92 · Fruit Trees F1=0.89 Large support + spectrally distinct July NDVI peak.

Oil & Fiber anomaly — Prec=1.00 Rec=0.20 Over-conservative. 80% of instances misclassified as Cereals. Overlapping SWIR-1 at peak season.

Herbs & Spices F1=0.00 n=5 test samples. Statistically meaningless — no valid inference possible.

03 DATA CONTEXT

Phenology & benchmarking

RE_NDVI_Jul = #1 feature Jul+Aug = 33% of total Gini gain. Peak window where crop canopies are most spectrally distinct. Red-edge separates chlorophyll concentration — unavailable in SAR.

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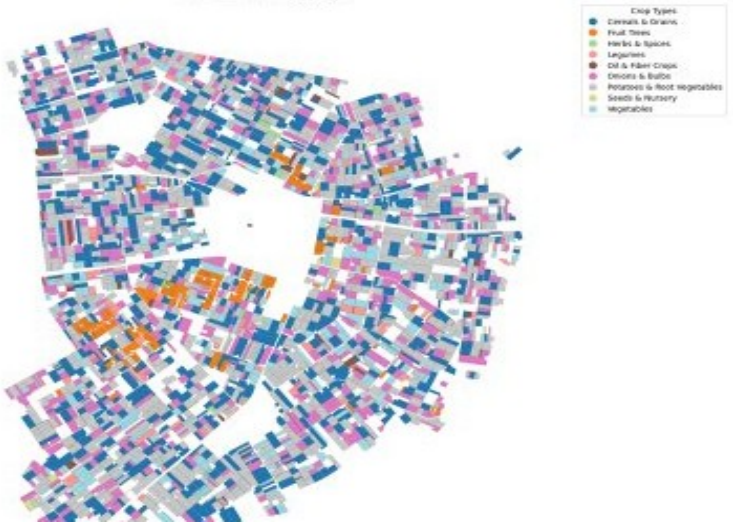
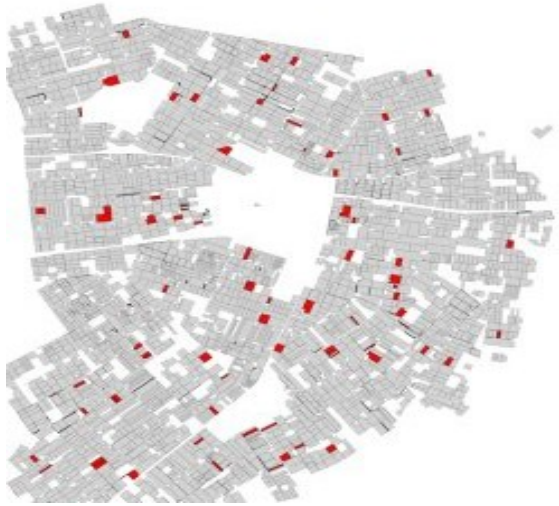
Error Analysis — What is Driving Misclassification?

Full Classified Crop Map — Dronten Test Set



Misclassification Map (red = wrong prediction)



1 Spatial resolution — NOT the cause

Only 4.3% of errors from sub-pixel parcels. 87.6% of errors occur on fields large enough for 20m. Sensor is not the problem.

87.6%
errors on large parcels

2 Neighbouring fields — NOT the cause

Correct and incorrect parcels show identical neighbour heterogeneity — just 1% difference. Being surrounded by mixed crops does not increase error rate.

1%
neighbour diversity diff.

3 Spectral similarity — contributing factor

Oil & Fiber, Legumes, Vegetables share near-identical NDVI profiles. Oil & Fiber Prec=1.00 but Rec=0.20 — 80% of instances look too similar to Cereals spectrally to distinguish.

Rec=0.20
Oil & Fiber missed

4 Class imbalance — PRIMARY driver

Potatoes (222) + Cereals (187) + Onions (111) = 65.7% of all 3,143 errors — simply from dominating training data. Rare classes fail from too few examples, not spectral confusion.

65.7%
errors from top 3 classes